Worksheet 6.3 — Proving Trigonometric Identities

Grade 12 Advanced Functions (MHF4U) · Unit 6: Trigonometric Identities and Equations

The Big Ideas

Transform one side using the Pythagorean, quotient, and reciprocal identities until it matches the other.

- $\sin^2\theta + \cos^2\theta = 1$; $1 + \tan^2\theta = \sec^2\theta$; $1 + \cot^2\theta = \csc^2\theta$.
- $\tan\theta = \frac{\sin\theta}{\cos\theta}$, $\cot\theta = \frac{\cos\theta}{\sin\theta}$.
- Start with the harder side; rewrite in sines and cosines.

Core Concepts

What a proof shows

An **identity** holds for all valid angles; a proof transforms one side into the other using known identities and algebra.

A steady strategy

Start from the more complex side, rewrite everything in sines and cosines, and lean on the Pythagorean and compound identities.

Not an equation

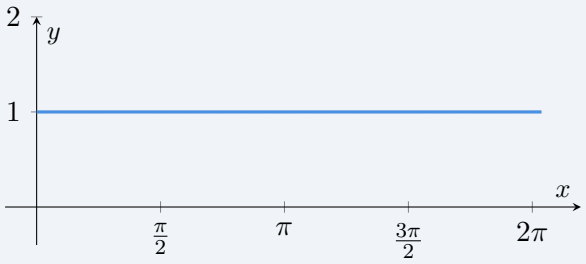
You never move terms across the equals sign each step simply rewrites one side until the two match.

Worked Examples

Example 1 — Pythagorean

Prove $\dfrac{\sin^2 x}{1-\cos x} = 1 + \cos x$.

Solution. $\sin^2 x = 1 - \cos^2 x$ (the sum below is always 1): $\dfrac{(1-\cos x)(1+\cos x)}{1-\cos x} = 1 + \cos x$. ✓



Example 2 — Quotient

Prove $\tan x \cos x = \sin x$.

Solution. $\dfrac{\sin x}{\cos x} \cdot \cos x = \sin x$. ✓

Example 3 — Reciprocal ratios

Prove $\dfrac{\tan x}{\sin x} = \sec x$.

Solution. $\dfrac{\sin x / \cos x}{\sin x} = \dfrac{1}{\cos x} = \sec x$. ✓

Example 4 — Build to a known one

Prove $\cot x \sec x = \csc x$.

Solution. $\dfrac{\cos x}{\sin x} \cdot \dfrac{1}{\cos x} = \dfrac{1}{\sin x} = \csc x$. ✓

Example 5 — Pythagorean form

Prove $\cos^2 x\,(1 + \tan^2 x) = 1$.

Solution. $1 + \tan^2 x = \sec^2 x$, so $\cos^2 x \cdot \dfrac{1}{\cos^2 x} = 1$. ✓

Example 6 — Difference of squares

Prove $(1 - \sin x)(1 + \sin x) = \cos^2 x$.

Solution. $1 - \sin^2 x = \cos^2 x$. ✓

Example 7 — Quotient

Prove $\cot x \sin x = \cos x$.

Solution. $\dfrac{\cos x}{\sin x} \cdot \sin x = \cos x$. ✓

Example 8 — Quotient

Prove $\dfrac{\cos x}{\cot x} = \sin x$.

Solution. $\cos x \cdot \dfrac{\sin x}{\cos x} = \sin x$. ✓

Example 9 — Combine

Prove $\sec x - \cos x = \sin x \tan x$.

Solution. $\dfrac{1}{\cos x} - \cos x = \dfrac{1 - \cos^2 x}{\cos x} = \dfrac{\sin^2 x}{\cos x} = \sin x \cdot \dfrac{\sin x}{\cos x}$. ✓

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Prove $\cos^2 x - 1 = -\sin^2 x$.

Question 2

Prove $\cot x \tan x = 1$.

Question 3

Prove $\dfrac{\cot x}{\csc x} = \cos x$.

Question 4

Prove $\dfrac{\sec x}{\tan x} = \csc x$.

Question 5

Prove $\sin^2 x \left(1 + \cot^2 x\right) = 1$.

Question 6

Prove $\dfrac{\sin x}{\tan x} = \cos x$.

Question 7

Prove $\dfrac{\csc x}{\cot x} = \sec x$.

Question 8

Prove $\sin x \sec x = \tan x$.

Question 9

Prove $\cos x \csc x = \cot x$.

Question 10

Prove $\sec^2 x - \tan^2 x = 1$.

Question 11

Prove $\dfrac{1-\cos^2 x}{\sin x} = \sin x$.

Question 12

Prove $\dfrac{\cos x}{1-\sin^2 x} = \sec x$.

Question 13 — Challenge

Prove $\dfrac{\sin x}{1+\cos x} + \dfrac{1+\cos x}{\sin x} = \dfrac{2}{\sin x}$.

Question Recap & Answer Key

Question Recap

1. Prove $\cos^2 x - 1 = -\sin^2 x$.
2. Prove $\cot x \tan x = 1$.
3. Prove $\dfrac{\cot x}{\csc x} = \cos x$.
4. Prove $\dfrac{\sec x}{\tan x} = \csc x$.
5. Prove $\sin^2 x\,(1 + \cot^2 x) = 1$.
6. Prove $\dfrac{\sin x}{\tan x} = \cos x$.
7. Prove $\dfrac{\csc x}{\cot x} = \sec x$.
8. Prove $\sin x \sec x = \tan x$.
9. Prove $\cos x \csc x = \cot x$.
10. Prove $\sec^2 x - \tan^2 x = 1$.
11. Prove $\dfrac{1 - \cos^2 x}{\sin x} = \sin x$.
12. Prove $\dfrac{\cos x}{1 - \sin^2 x} = \sec x$.
13. Prove $\dfrac{\sin x}{1 + \cos x} + \dfrac{1 + \cos x}{\sin x} = \dfrac{2}{\sin x}$.

✓ Answer Key

1. $-(1 - \cos^2 x) = -\sin^2 x$
2. $\frac{\cos x}{\sin x} \cdot \frac{\sin x}{\cos x} = 1$
3. $\frac{\cos x / \sin x}{1/\sin x} = \cos x$
4. $\frac{1/\cos x}{\sin x/\cos x} = \frac{1}{\sin x}$
5. $\sin^2 x \csc^2 x = 1$
6. $\sin x \cdot \frac{\cos x}{\sin x} = \cos x$
7. $\frac{1/\sin x}{\cos x/\sin x} = \frac{1}{\cos x}$
8. $\sin x \cdot \frac{1}{\cos x} = \tan x$
9. $\cos x \cdot \frac{1}{\sin x} = \cot x$
10. from $1 + \tan^2 x = \sec^2 x$
11. $\frac{\sin^2 x}{\sin x} = \sin x$
12. $\frac{\cos x}{\cos^2 x} = \frac{1}{\cos x}$
13. common denom $\rightarrow \frac{\sin^2 x + (1+\cos x)^2}{\sin x(1+\cos x)} = \frac{2}{\sin x}$